

Module

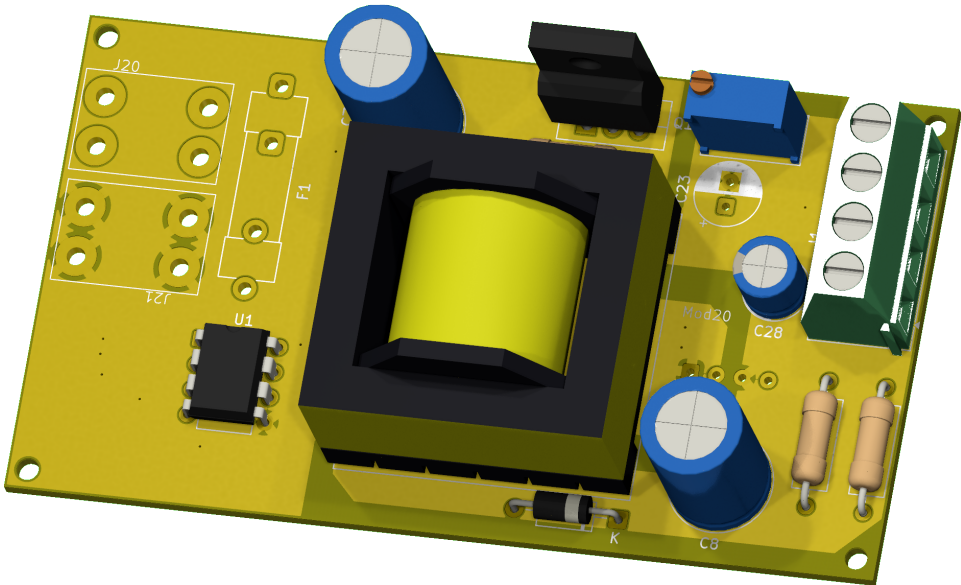
Rev 1.0.0+ (Unreleased)

Variant: PRELIMINARY

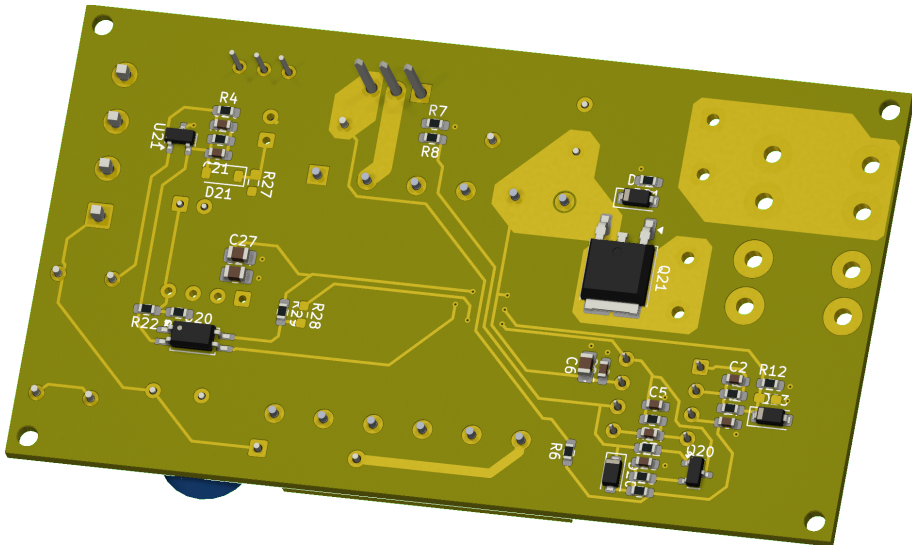
2026-01-28

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TOP VIEW



BOTTOM VIEW



DRAFT - Early concept stage, schematic drafted.

PRELIMINARY - Minor schematic changes are expected, ready for layout planning.

CHECKED - DRC completed, schematic and layout ready for review.

RELEASED - Design ready for manufacturing.

NOTE STYLING CONVENTION

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

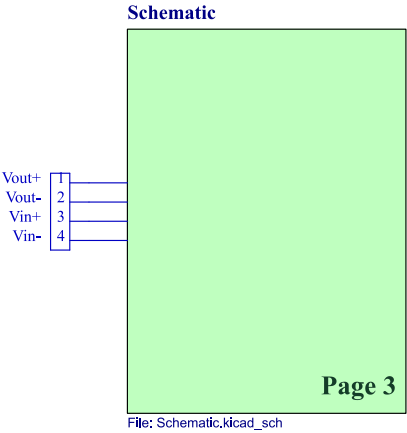
Example text for critical layout guidelines.

Date: 28-Jan-2026

	Comments:	Company: FR	Variant: PRELIMINARY	Git Hash: 120601b
	Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title:	File Name: Isolated_12to450V.kicad_sch	Designer: FR	Date: 2025-12-20
	Sheet Path: /	Reviewer: NA	Size: A3	Revision: 1.0.0+ (Unreleased)
				Sheet: 1 of 4

[2] Block Diagram

MODULE INTERFACE:
Vin: 9 V to 24 V according
to AliExpress product page
Vout: 200 V to 450 V,
adjustable through VR1

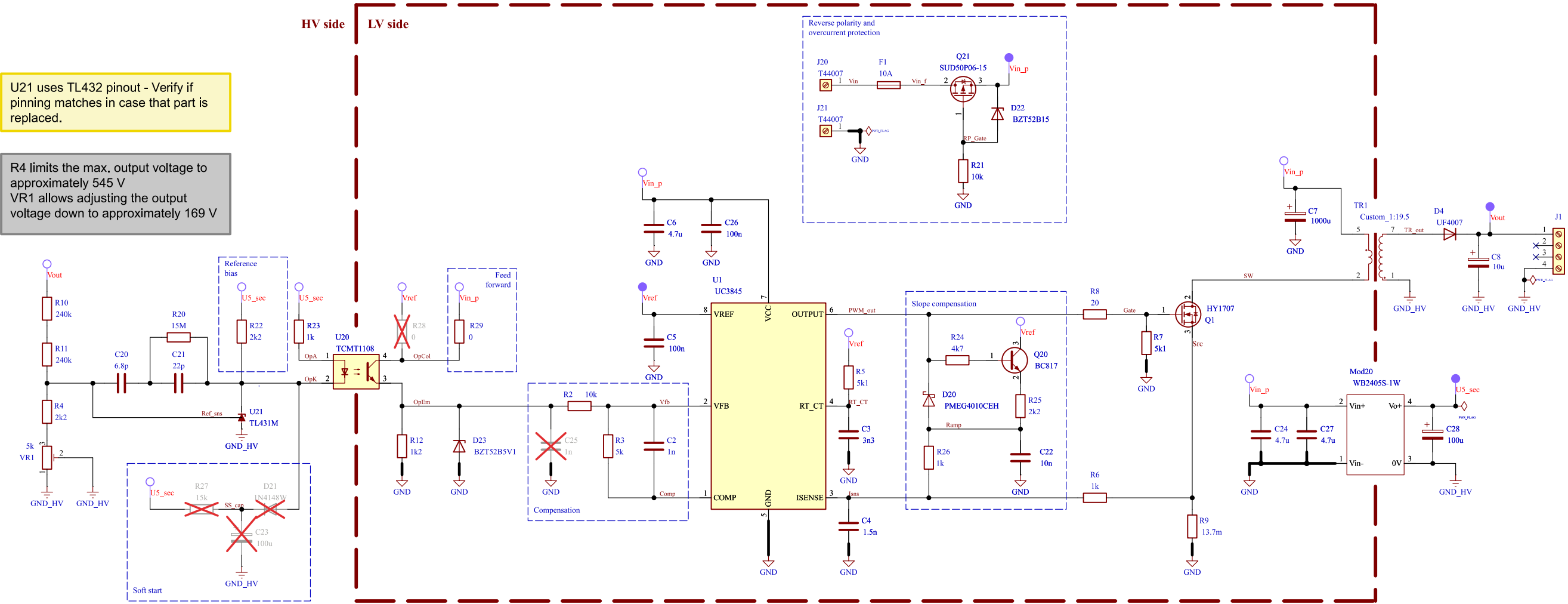


	Comments: AliExpress module example (link)		Company: FR		Variant: PRELIMINARY	Git Hash: 120601b
	Board Name: Module		Project Name: Isolated step-up 12:450 V			
	Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: FR	Date: 2025-12-20	Revision: 1.0.0+ (Unreleased)	
	Sheet Path: /Block Diagram/			Reviewer: NA	Size: A4	Sheet: 2 of 4

[3] Schematic

U21 uses TL432 pinout - Verify if pinning matches in case that part is replaced.

R4 limits the max. output voltage to approximately 545 V
VR1 allows adjusting the output voltage down to approximately 169 V



Estimated specs for unidentified / custom parts:
- R9: ~13.7 mΩ (110 mV @ 8.04 A)
- TR1: ~1:19.5 turns ratio, 3.54:1453 uH

- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

Comments:	Company: FR		Variant: PRELIMINARY	Git Hash: 120601b
	Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: Schematic	File Name: Schematic.kicad_sch	Designer: FR	Date: 2025-12-20
Sheet Path: /Block Diagram/Schematic/		Reviewer: NA	Size: A3	Revision: 1.0.0+ (Unreleased)
			Sheet: 3 of 4	

[4] Revision History

Version 1.0.0 - 2026-01-24

Cloned non-isolated design revision 1.1.0 as a working base.

Updated R10 and R11 to 240 kΩ (cloned design incorrectly used 140 kΩ).

Updated U1 to UC3845 (cloned design incorrectly used UC3843).

Reworked feedback loop to include opto-coupler while keeping the same reference on both sides to facilitate testing / reuse of original board.

Added two options of slope compensation / ramp injection (from U1 RT_CT pin and from U1 OUTPUT pin).

Adjusted R2 to 10 kΩ, R3 to 5 kΩ and C2 to 1 nF to improve behaviour (output swaps from bursts to 0 % duty periodically).

Added soft-start (D21, R27 and C23).

Increased C5 to 100 nF.

Adjusted C4 to 1.5 nF and R6 to 1 kΩ (similar time constant) to facilitate ramp injection.

Updated parts placement where suitable to reuse existing layout and added an auxiliary perf-board to accomodate the new parts.

Version 1.0.1 - 2025-12-21

	Comments:	Company: FR		Variant: PRELIMINARY	Git Hash: 120601b
		Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: FR	Date: 2025-12-20	Revision: 1.0.0+ (Unreleased)
	Sheet Path: /Revision History/		Reviewer: NA	Size: A4	Sheet: 4 of 4